

**SOM**

Semester: July 2023 – Oct 2023			Duration :
Examination: In-Semester Examination			
Maximum Marks:	Programme code: 01	Class: TY	Semester: V (SVU 2020)
Programme: B.Tech. Computer Engineering		Name of the department: COMP	
Name of the Constituent College:			
K. J. Somaiya College of Engineering		Name of the Course: Soft Computing	
Course Code: 116U01E514			

Question No.		Max. Marks	CO Mapped	BT Level																				
Q1	Explain Delta Learning Rule wrt the following terms: Mean square error Gradient descent Weight updation formula OR Implement AND function using Perceptron networks for bipolar inputs and targets.	10	CO2 CO2	UN / RE / AP																				
Q2	Use Adaline network to train following function with bipolar inputs and targets. Perform 1 epoch of training. Calculate total mean square error. <table border="1"><tr><td>x1</td><td>x2</td><td>1</td><td>t</td></tr><tr><td>1</td><td>1</td><td>1</td><td>-1</td></tr><tr><td>1</td><td>-1</td><td>1</td><td>1</td></tr><tr><td>-1</td><td>1</td><td>1</td><td>-1</td></tr><tr><td>-1</td><td>-1</td><td>1</td><td>-1</td></tr></table>	x1	x2	1	t	1	1	1	-1	1	-1	1	1	-1	1	1	-1	-1	-1	1	-1	10	CO2	AP
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-1	1	1	-1																					
-1	-1	1	-1																					
Q3	Attempt any 2: a. Explain Linearly separable and inseparable problems. b. What do you understand by Training and Testing of Neural Network? Differentiate between Supervised and unsupervised learning Algorithms. c. Construct McCulloch-Pitts neuron model for AND function.(Use binary data)	10	CO1,CO2	RE/ UN																				